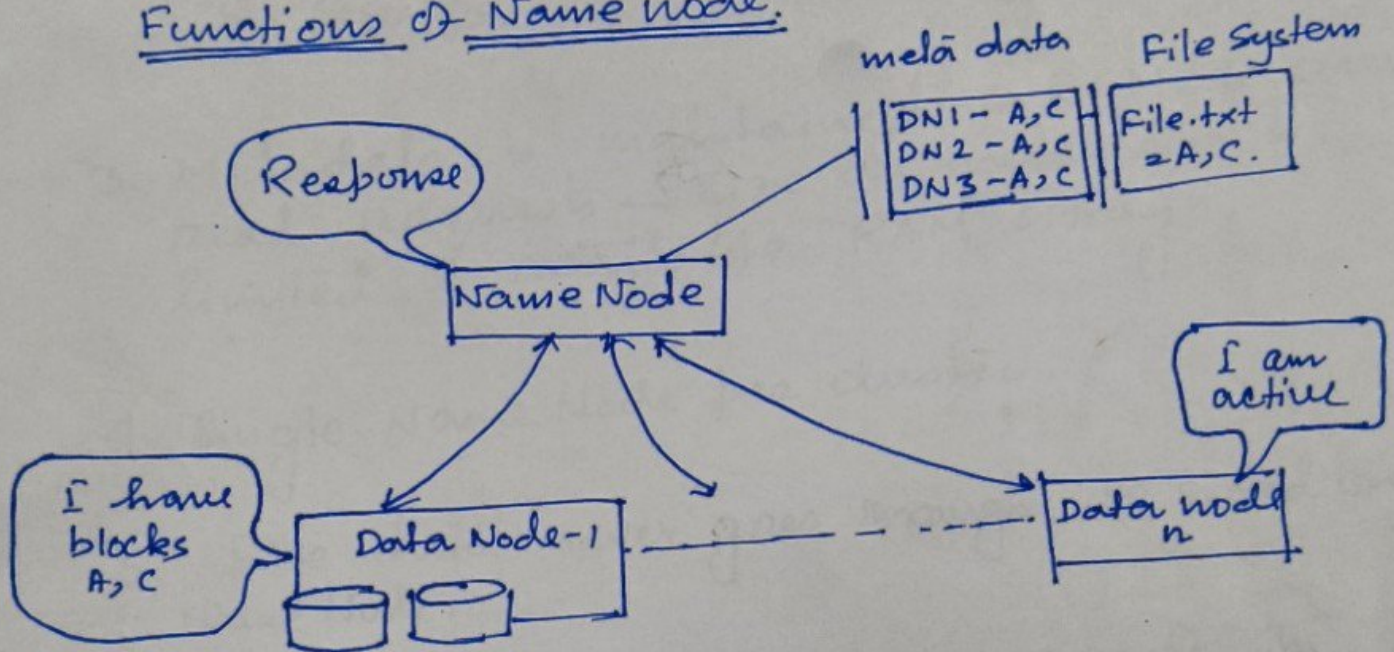


9

Name Node:

Name node is the master of HDFS that maintains and manages the blocks present on the data nodes. Name node is a very high availability - availability server that manages the file system namespace and controls access to files by system.

Functions of Name node:



- Data nodes send heartbeats.
- Every 10 heartbeats is a Block Report.
- Name node builds meta data from block reports.
- TCP - every 3 seconds
- If, name node is down the Hadoop HDFS becomes inoperative.

Functions of a Name Node:

1. Manages the File system namespace.
2. stores all metadata in memory.
 1. File name
 2. Locations of all blocks for a file.
 3. permissions.
 4. Owner
 5. Groups.
3. Metadata is maintained in RAM to serve read requests - Size of metadata is limited to available RAM storage.
4. Single Name Node per cluster.
5. File data never goes through or stored on Name Node.
6. It directs the Data Nodes to execute the low-level I/O operations.
7. It keeps a record of how the files in HDFS are divided into blocks, in which nodes these blocks are stored and by and large the NameNode manages cluster configuration.
8. It maps a filename to a set of blocks and maps a block to the DataNodes where it is located.

9. It records the metadata of all the files stored in the cluster, e.g. the location, the size of the files, permission hierarchy etc.

10. With the help of a transactional log i.e. the Edit log it records each and every updation taking place.

11. Finally it takes the responsibility of to coordinate the activities of all the DataNodes

12. It is mandatory that the Name Node should be formatted (Hadoop Name-node format).

Data Nodes:

Data Nodes are slave nodes in HDFS. The functions of Data Nodes are:

1. storage for file blocks.
2. Different blocks from the same files are stored in different nodes. DataNodes.
3. Replication factor is configurable.
4. Size of the blocks are configurable.
5. DataNodes communicate with the Name Node and other DataNodes.
6. DataNodes does not need to be explicitly formatted as does the NameNode.
7. A datanode may contain multiple disks.
 - i) They are controlled by dfs.data.dir.
 - ii) Blocks are written to each in round-robin fashion.
 - iii) DataNode perform the low level file read and write requests from the file system's client.

Job Tracker

The Job Tracker daemon is the liaison between the client's application and Hadoop. Once the job is submitted to the cluster, the job tracker determines the execution plan by determining which files to process, assigns nodes to different tasks and monitors all tasks as they are running. If instantaneously a task fails, the job tracker automatically re-launches the task possibly on a different node upto a predefined limit of retries.

Per Hadoop cluster there is only one job tracker. It is typically run on a server as a master node of the cluster.

Task Tracker

Each Task Tracker is responsible for executing the individual tasks that the job tracker assigns. Although there is a single Task Tracker per slave node, each Task Tracker can spawn multiple Java virtual machines (JVM) to handle many map or reduce tasks in parallel. One responsibility of the Task Tracker is to constantly communicate with the job tracker. If the job tracker fails to receive a heart beat from a Task Tracker within a specified amount of time, it will assume that the Task Tracker has crashed and will resubmit the job corresponding task to other nodes in the cluster.

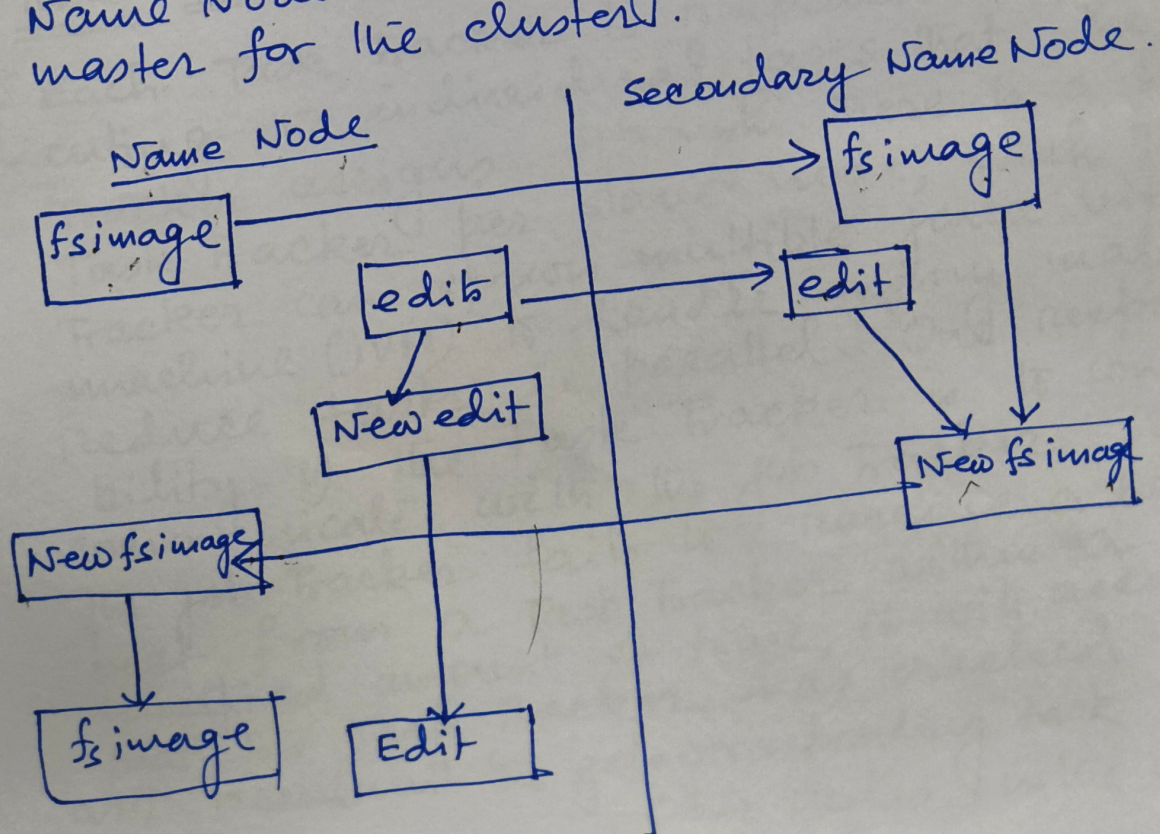
~~It is~~. There are two associated files:

fsimage: An image of the file system which keeps the information of all the files at the starting of the NameNode.

Edit logs: A series of modifications done to the file system after starting the NameNode.

The secondary NameNode downloads the Edit logs from the NameNode at regular intervals and applies it to fsimage. The new fsimage is copied back to the NameNode, which is used whenever the NameNode is started the next.

If the NameNode fails, the entire HDFS goes down, in this case we have to start secondary NameNode manually which acts like the master for the cluster.



8. They are also responsible for creating blocks and replicating the same based on the decision taken by the NameNode.
9. They regularly send a report on all the blocks present in the cluster to the NameNode.
10. Datanodes also enables pipelining of data.
11. They forward data to other specified DataNodes.
12. DataNodes send heart beats to the NameNode every 3 seconds to report the overall health of HDFS.

Secondary NameNode

In the HDFS architecture, the ~~name-ns~~ secondary NameNode gives an impression that it is a substitute of the NameNode. But in reality it performs a supporting job to the NameNode. It constantly reads all the files systems and metadata from the RAM of the NameNode and writes it into the hard disk or the file system.